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ELECTRICITY USE REVIEW

Harmony @Huailan Ecolodge: how your power is used

Solar + battery system, 10 kWp panels, 10 kW inverter · Data: 1 October – 1 November 2026, readings every 5 minutes

The short version

- Your solar system works well. No faults or alarms were found.
- You used **more electricity this year** than last year: about 14 kWh a day in October (+12%). You also bought more from the grid (+52%).
- Most of what you buy from the grid is caused by **two settings** that can be fixed for free. The battery stops at 40% instead of 25%, and it still refills from the grid every morning.
- About once a day, something very large switches on, around **11–12 kW** for a few minutes, usually early morning or late evening. It is most likely water heating.
- The busy cool season is starting. Last year November and December were your highest months.

1. How much electricity you use

In 2026 the lodge has used between 310 and 490 kWh a month. Since June, every month except July has been higher than the same month last year. Most electricity comes from the sun, either directly or stored in the battery. The rest is bought from the grid (red).

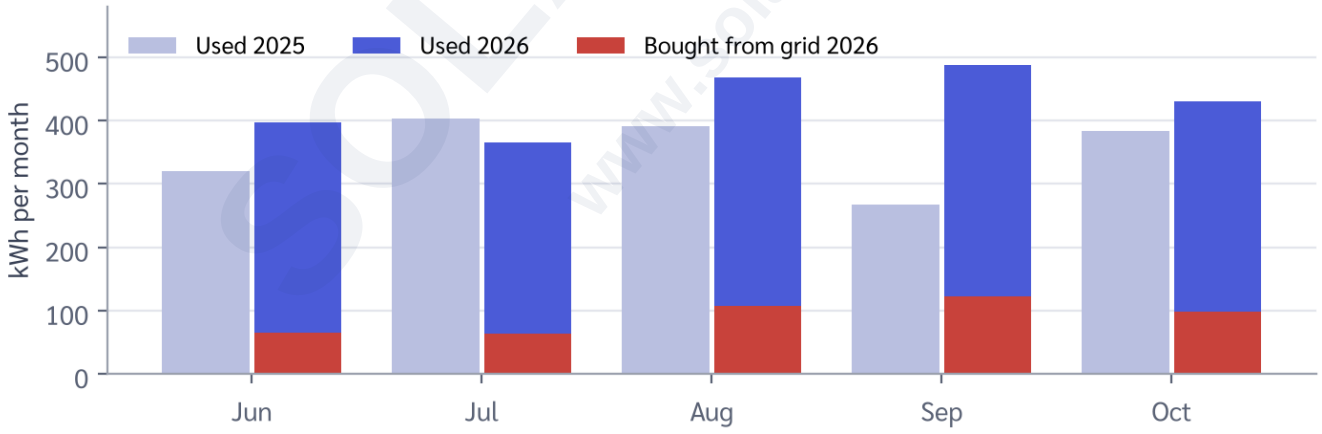


Figure 1. Monthly electricity used, 2025 vs 2026, and how much was bought from the grid in 2026.

Day by day, use changes a lot with the number of guests. Quiet days use only 2–5 kWh. Busy days such as 17–20 October use 20–27 kWh. Busy days are when most grid power is bought.

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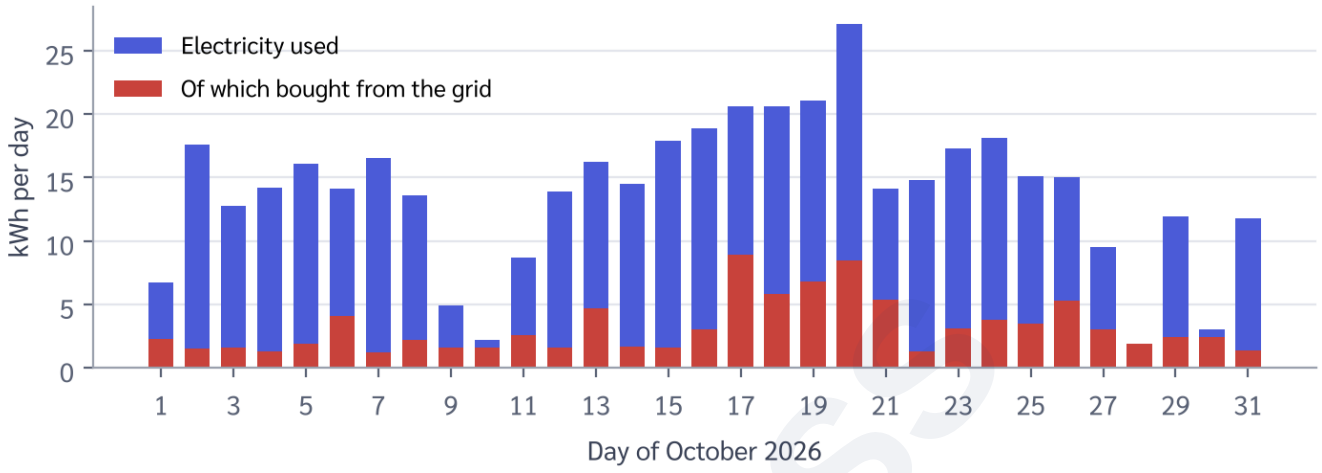


Figure 2. Electricity used each day in October (blue) and the part bought from the grid (red).

2. A typical day

The chart shows an average October day, hour by hour.

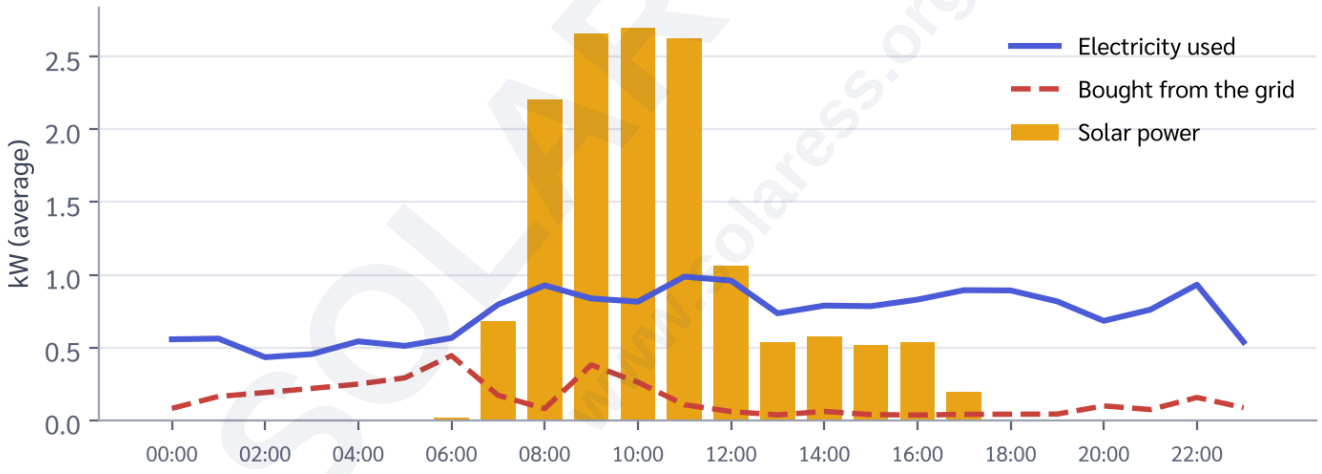


Figure 3. Average power through the day in October.

- **Night (01:00–06:00):** the lodge never goes below about **0.45–0.55 kW**. This "always-on" use (fridges, freezers, pumps, Wi-Fi, outdoor lights) adds up to about 11 kWh a day, roughly 80% of everything you use.
- **Morning (07:00–12:00) and evening (17:00–19:00, 22:00):** the busiest times, around 0.8–1.0 kW on average.
- **Solar:** the panels produce up to about 2.7 kW in the late morning. After about 12:00 the battery is full, so the system holds the panels back. Roughly 6–8 kWh of free solar a day is not used.

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3. Power peaks: when, how big, how long

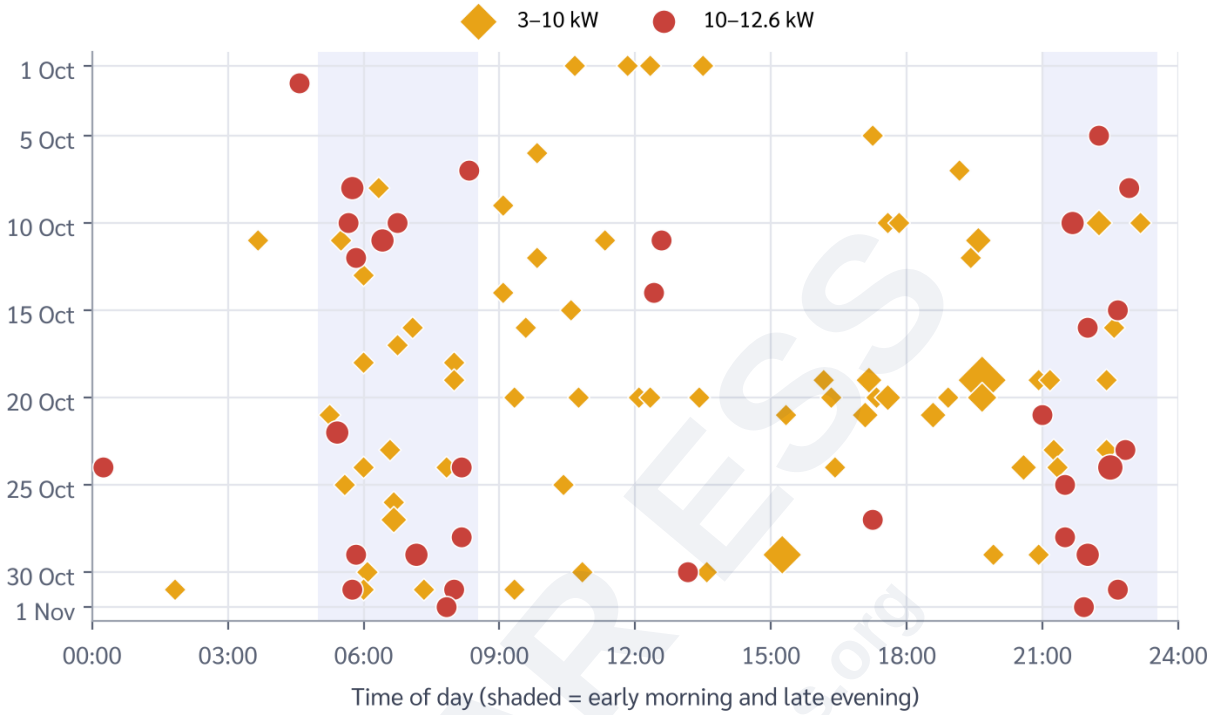


Figure 4. Every time the lodge used more than 3 kW. Red = 10–12.6 kW, orange = 3–10 kW. Bigger marks lasted longer.

Very large peaks (10–12.6 kW), about once a day. There were 34 in 32 days, almost always between 05:25–08:20 or 21:00–23:00. Each one lasts only a few minutes; usually only one 5-minute reading shows it. Every time it is the same size and shape: about 5.5–6 kW on two of your three electrical phases at once. That points to **one large two-phase appliance**, most likely a water heater or hot-water boiler. It could also be two heaters switching on together. Your inverter can give about 8.5 kW, so the grid supplies the remaining 3–4 kW during these peaks.

Smaller morning peak (about 4 kW), most days around 06:00. This is a single-phase appliance (for example a water heater, kettle or pump) running for a few minutes. Because the battery is held at 40% at that time, **all** of this power comes from the grid.

Long busy periods. The longest was on 19 October, 19:40–20:40: about one hour at 4–6 kW. 20 October was the busiest day of the month (27 kWh).

Good news: almost all lodge circuits are on the normal grid side, not on the inverter's backup outlet (backup load is only about 30–130 W). So these peaks cannot overload the inverter. It also means that during a power cut, only the small backup circuit stays on.

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4. Why you still buy power from the grid

We checked the inverter settings. The battery timetable (6 time slots across the day) is set to:

Time slot	01–05	05–09	09–13	13–17	17–21	21–01
Lowest battery level now	40%	40%	80%	80%	40%	40%
Charge battery from grid now	Yes	Yes	Yes	Yes	Yes	Yes
Recommended lowest level	25%	25%	25%	25%	25%	25%
Recommended grid charging	No	No	No	No	No	No

- Because the lowest level is 40%, the battery stops helping at night while 15% of it, about 2 kWh, is still unused. The lodge then buys from the grid.
- Because grid charging is on, and the level is 80% from 09:00 to 17:00, the system **buys grid power in the morning to refill the battery**, just before the sun could do it for free.

5. What we recommend

- **Change the battery timetable (free).** Set all six slots to 25% and turn grid charging off. Solar ESS will do this remotely with your agreement. We expect it to remove most night-time and morning grid buying on normal days.
- **Run big loads in the middle of the day (free).** Use laundry, ironing, pumps and water heating between 11:00 and 16:00, when the battery is full and solar is otherwise wasted.
- **Find the large heater.** On our next visit we will identify the 11–12 kW appliance. A heat-pump water heater or solar hot water would cut both the peaks and the evening grid use. Using one heater at a time also helps.
- **Check what stays on all night.** Old fridges or freezers, pumps and lights left on can be 0.45 kW around the clock. A small cut here saves every day of the year.
- **Decide about a bigger battery later.** After the settings change, we will watch November–December (the busy season). If you still buy a lot of power on busy nights, more battery storage would use the solar that is being wasted today.

Technical notes (for your electrician)

Inverter: three-phase LV hybrid, 10 kW rated; lithium battery, 300 Ah at 48 V. Large peaks appear on L2 and L3 together (about 5.5–6.3 kW each), with L1 almost unchanged. During these peaks, L2 voltage rose to about 248–252 V and L3 fell to about 218–223 V. That is within limits but worth checking (neutral and supply connections) on the next visit. Battery settings: Shutdown 12%, Low 25%, Restart 50%; the time-of-use slot SOC values (40/40/80/80/40/40) override the 25% low level. Update: on 2 Nov 2026 the settings were changed to 25% in every slot with grid charging off (see our field note of the same day). Data source: the online monitoring platform of the inverter.

Questions? Call Solar ESS on 063 151 6694 or 089 632 5829, or LINE: solaress1.

Prepared by Solar ESS Working Group

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